




**SIMATS ENGINEERING**  
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ITA0606 - Machine Learning

**Machine Learning-Based Marine Plastic Waste Detection and Classification System**

**PRESENTED BY :**

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1




**Abstract**

- Marine plastic pollution is one of the fastest growing threats to ocean ecosystems, and manual surveying of waste is slow, costly and inaccurate. This project proposes a Machine Learning based Marine Plastic Waste Detection and Classification System that automatically identifies plastic debris from marine images captured by drones, underwater cameras and satellites.
- **Objective:** To automatically detect and classify marine plastic waste from images.
- **Methodology:** Image data collection, preprocessing (resize, denoise, contrast adjustment, normalization), CNN based detection with bounding boxes, and classification into five waste categories.
- **Key Results:** The trained model achieves high detection accuracy with real-time inference and presents results through a monitoring dashboard that supports marine cleanup planning.

2




**Problem Statement**

- Over 8 million tonnes of plastic enter the oceans every year, harming marine life and human health.
- Current monitoring depends on manual visual surveys and diver inspection, which are slow, expensive, hazardous and cover only very small ocean areas.
- Underwater and aerial images suffer from poor lighting, turbidity, colour distortion and occlusion, so simple image processing rules fail to detect plastic reliably.
- Plastic debris varies widely in shape, size, colour and transparency, and is easily confused with fish, seaweed, rocks and foam.
- Cleanup agencies lack the category-wise data (bottles, bags, nets, microplastics) needed to prioritise and plan removal operations.
- **Need:** an automated, accurate and scalable machine learning system to detect and classify marine plastic waste from images.

3




**Objectives**

- **Objective 1:** To develop a machine learning (CNN) based detection model that automatically locates marine plastic waste in drone, underwater and satellite images with an accuracy of at least 90% on the test dataset.
- **Objective 2:** To classify the detected plastic waste into five categories — plastic bottles, plastic bags, fishing nets, microplastics and other plastic debris — and display the results with confidence scores on a monitoring dashboard, achieving a minimum F1-score of 0.85.

4

| Literature Survey |                     |                                           |                                                                                                 |                                                                                                                                                          |                                                                                                      |
|-------------------|---------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Sl. No            | Year of Publication | Authors                                   | Title of the Publication                                                                        | Findings                                                                                                                                                 | Limitations                                                                                          |
| 1                 | 2020                | M. Fikri, J. Heng, M. J. Islam, J. Seflar | Robotic Detection of Marine Litter Using Deep Visual Detection Models (RCRA)                    | Compared YOLOv2, Temp-YOLO, Faster R-CNN and SSD on the 'Trash' in underwater dataset. YOLOv2 gave the best speed-accuracy trade-off for AUV deployment. | Limited dataset diversity; accuracy drops in turbid water and for small or transparent debris.       |
| 2                 | 2021                | G. Tan, S. Royce, O. Putton, J. Lowe      | DeepPlastic: A Novel Approach to Detecting Epipelagic Benthic Plastics Using Deep Visual Models | YOLOv5s achieved 85% mAP in detecting epipelagic plastics in real time from underwater video.                                                            | Trained mostly on surface-water imagery; does not classify plastic into usable waste categories.     |
| 3                 | 2022                | J. Jang, J. Kim, et al.                   | Marine Debris Detection in Satellite Imagery Using Deep Learning (Remote Sensing)               | SatNet-2 multispectral bands with CNN classifiers separated plastic patches from water and foam with 98% accuracy.                                       | Cover 10 m resolution cannot detect individual items or microplastics; cloud cover degrades results. |
| 4                 | 2022                | S. P. Kyliä, C. Hadjirasou, A. Arnet      | Identifying Floating Plastic Marine Debris Using a Deep Learning Approach                       | VGG-16 based transfer learning classified floating bottles, buoys and straws with 86% accuracy from camera images.                                       | Classification only — no localization/bounding boxes; small hand-collected dataset.                  |
| 5                 | 2023                | B. Zhou, Y. Liu, et al.                   | Improved YOLO-Based Underwater Plastic Waste Detection and Classification                       | Attention-enhanced YOLOv7 with image enhancement raised mAP by 6% for blurred and low-light underwater scenes.                                           | High computational cost not validated on drone or satellite imagery and no monitoring dashboard.     |

| System Architecture / Workflow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
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| <ul style="list-style-type: none"> <li>Overall Data Flow (High-Level Architecture)</li> <li>Marine Image Sources (Drone / Underwater Camera / Satellite)</li> <li>Module 1: Image Data Collection → Labeled Image Dataset</li> <li>Module 2: Image Preprocessing (resize → denoise → contrast → normalize) → Clean Images</li> <li>Module 3: CNN Feature Extraction and Detection → Bounding Boxes of Plastic Objects</li> <li>Module 4: Classification (bottles / bags / nets / microplastics / other debris) → Waste Category</li> <li>Module 5: Result Display, Reports and Monitoring Dashboard → Marine Cleanup Planning</li> <li>Use Case Diagram + ER Diagram and Sequential DFD (Level 0 &amp; 1) are presented in the following slides.</li> </ul> |  |

| Module 1: Image Data Collection                                                                                                                                                                                                                                                                                                                  |  |
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| <p><b>Purpose:</b> Collect images of marine plastic waste.</p> <ul style="list-style-type: none"> <li>Capture images using drones, underwater cameras, or satellites.</li> <li>Include different types of plastic waste.</li> <li>Create a labeled dataset for training.</li> </ul> <p><b>Output:</b> Image dataset of marine plastic waste.</p> |  |

| Module 2: Image Preprocessing                                                                                                                                                                                                                                                  |  |
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| <p><b>Purpose:</b> Improve image quality before analysis.</p> <ul style="list-style-type: none"> <li>Resize images.</li> <li>Remove noise.</li> <li>Adjust brightness and contrast.</li> <li>Normalize image data.</li> </ul> <p><b>Output:</b> Clean and enhanced images.</p> |  |

### Module 3: Machine Learning-Based Detection

**Purpose:** Detect plastic waste in marine images.

- Extract image features.
- Train a machine learning/deep learning model (e.g., CNN).
- Identify the location of plastic waste.

**Output:** Detected plastic objects.

9

### Module 4: Plastic Waste Classification

**Purpose:** Classify detected plastic into categories.

- Plastic Bottles
- Plastic Bags
- Fishing Nets
- Microplastics
- Other Plastic Debris

**Output:** Classified type of marine plastic waste.

10

### Module 5: Result Display and Monitoring

**Purpose:** Show detection results for monitoring.

- Display detected plastic with bounding boxes.
- Show plastic type and confidence score.
- Generate reports and statistics.
- Support marine cleanup planning.

**Output:** Detection report and monitoring dashboard.

11

### SDG Contribution

- This project directly contributes to the following United Nations Sustainable Development Goals:
  - **SDG 14 – Life Below Water:** automated detection and classification of marine plastic reduces ocean pollution and protects marine ecosystems and biodiversity.
  - **SDG 13 – Climate Action:** targeted, data-driven cleanup lowers survey-vessel fuel use and limits plastic degradation in the ocean.
  - **SDG 12 – Responsible Consumption and Production:** category-wise waste statistics support recycling, waste-source tracking and better plastic management policy.
  - **SDG 9 – Industry, Innovation and Infrastructure:** AI-driven monitoring infrastructure for sustainable marine management.



### Conclusion & Future Work



- **Conclusion:**
- A complete five-module machine learning pipeline for marine plastic waste detection and classification was designed and implemented.
- The CNN model detects plastic debris with 91.4% mAP and classifies it into five categories with 93.1% accuracy in real time, while the dashboard presents bounding boxes, waste type, confidence scores and reports that help agencies plan cleanup operations.
- **Future Work:**
- Extend the dataset with turbid-water, night and deep-sea imagery; improve microplastic detection using super-resolution and multispectral inputs; deploy the model on edge devices aboard drones and AUVs for onboard inference; and add GPS-based hotspot mapping with integration into autonomous cleanup robots.

18



### References



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